



The Diversity Crisis of Software Engineering for Artificial Intelligence

Bram Adams and Foutse Khomh

From the Editor

One factor enabling the current success of artificial intelligence in industry is the availability of high-quality machine learning toolkits. But what do we really know about those tools? In this column, researchers from the Polytechnique Montréal are concerned about how these tools are being built (and how they might negatively impact innovation and code quality). —*Tim Menzies*

ARTIFICIAL INTELLIGENCE (AI) is experiencing a “diversity crisis.”¹ Several reports^{1–3} have shown how the breakthrough of modern AI has not yet been able to improve on existing diversity challenges regarding gender, race, geography, and other factors, neither for the end users of those products nor the companies and organizations building them. Plenty of examples have surfaced in which biased data engineering practices or existing data sets led to incorrect, painful, or sometimes even harmful consequences for unassuming end users.⁴ The problem is that ruling out such biases is not

straightforward due to the sheer number of different bias types.⁵ To have a chance to eliminate as many biases as possible, most of the experts agree that the teams and organizations building AI products should be made more diverse.^{1–3} This harkens back to Linus’ law⁶ for open source development (“given enough eyeballs, all bugs are shallow”) but applied to the development process of AI products.

Unfortunately, current AI organizations and companies are not diverse. For example, AI NOW Institute’s West et al.¹ report on how work floor discrimination during hiring/promotion persists, or how AI companies still reason in terms of binary gender. Worse, current diversity

initiatives tend to introduce new biases, for example toward “white women.” Element AI’s Gagné et al.⁷ report how AI experts (self-identified via their LinkedIn profiles) primarily reside in North America, the United Kingdom, France, and Germany. Even in academia, which could be considered to be more progressive, they found fewer than one out of five authors at major AI conferences to be women (which coincidentally corresponds to the percentage of AI faculty estimated to be women).⁸ This was confirmed by Stathoulopoulos et al. for arXiv papers⁹ and by Simonite for publications (and the percentage of women employees) from R&D labs at major tech companies like Google, Microsoft, and IBM.²

In this column, which is based on our MSR 2018 publication,¹⁰ we argue that the “diversity crisis” of AI goes even further, and that it has other consequences that might hamper further evolution. This claim is based on an empirical study of the GitHub data of the 20 most popular company- and community-driven frameworks for machine learning (ML), and the Linked-In and Google Scholar profiles of the top contributors of these frameworks.¹⁰ Basically:

- the top ML software frameworks, while being open source, are primarily driven by companies and these frameworks are optimized for specific cloud offerings
- ML software projects converge toward highly specialized skills and roles

Both observations show additional sources of bias in modern AI software projects that, if ignored, might quickly widen the gap between the companies at the top of the tech chain and others. We also discuss a number of promising directions in which to proceed.

Methodology

To understand diversity in ML/AI software projects, we analyzed 598 core contributors of 20 top open source ML frameworks, out of a set of 104 ML-related open source frameworks.

The frameworks were selected by querying GitHub’s search application programming interface using technical terms related to ML, such as *machine learning*, *deep learning*, *statistical learning*, *neural network*, *supervised learning*, *unsupervised learning*, and *reinforcement learning* as well as known keywords such as *toolkit*, *tool*, *framework*, and *library*. The resulting search results were filtered manually

based on the license file, README, list of contributors (and affiliations), and GitHub’s built-in wiki, eventually arriving at a list of 104 projects.

Next, to approximate the number of developers who adopted these frameworks, we used GitHub’s built-in popularity measure, i.e., the number of people who starred the framework’s repository. (Our study also considered a second measure of adoption, i.e., 4,099 projects that used the ML frameworks. We refer to our paper for those results.¹⁰) By selecting the set of frameworks whose cumulative number of stars (adopters)

in the source code repository accounted for 90% or more of the total contributions. We then extracted GitHub, LinkedIn, and Google Scholar profiles of each contributor in this set.

Results

Company versus community driven

Our study shows that, although the development of open source ML frameworks was initially driven by the open source community, since 2013, the number of ML frameworks backed by companies has

To understand diversity in ML/AI software projects, we analyzed 598 core contributors of 20 top open source ML frameworks.

represents 80% of the total number of stars of the 104 ML frameworks, we obtained 29 popular frameworks. We then qualitatively analyzed these projects’ descriptions, their corresponding organizations, and their official websites on GitHub to determine whether they are mostly supported by a company or by the open source community. From this qualitative analysis, we selected the top 10 company-driven and the top 10 community-driven projects. These 20 projects represent 70% of the total number of stars attributed to the 104 ML frameworks on GitHub.

To identify the core contributors of these frameworks, we computed the total contribution of each contributor and retain the set of contributors whose total number of contributions

surpassed the number of community-driven ML frameworks. Furthermore, the number of adopters of ML frameworks supported by companies largely surpasses the number of adopters of community-driven ML frameworks, as shown in Figure 1.

The dominance of companies is much larger than suggested in Figure 1 because, apart from releasing their own ML frameworks, companies also actively contribute to community-driven ML frameworks by providing and/or supporting skilled professionals. Moreover, a deep analysis of the source code and dependencies of community-driven frameworks revealed that all community-driven frameworks that appeared since 2014 are in fact built on company-driven solutions, as shown in Figure 2. This increases

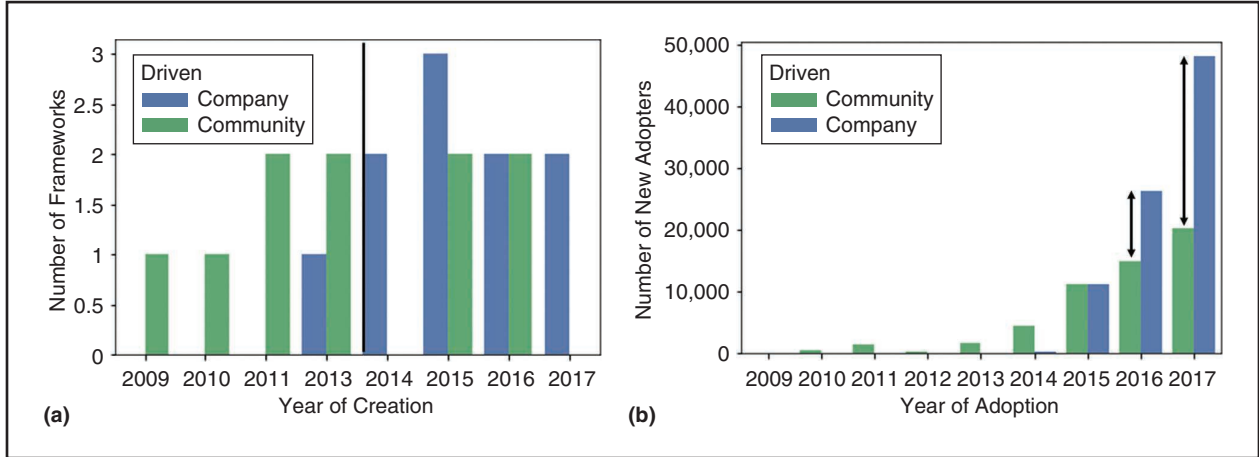


FIGURE 1. A comparison of company-versus community-driven ML frameworks: (a) number of frameworks for each type of project per year of creation and (b) number of adopters for each type of project per year of adoption.

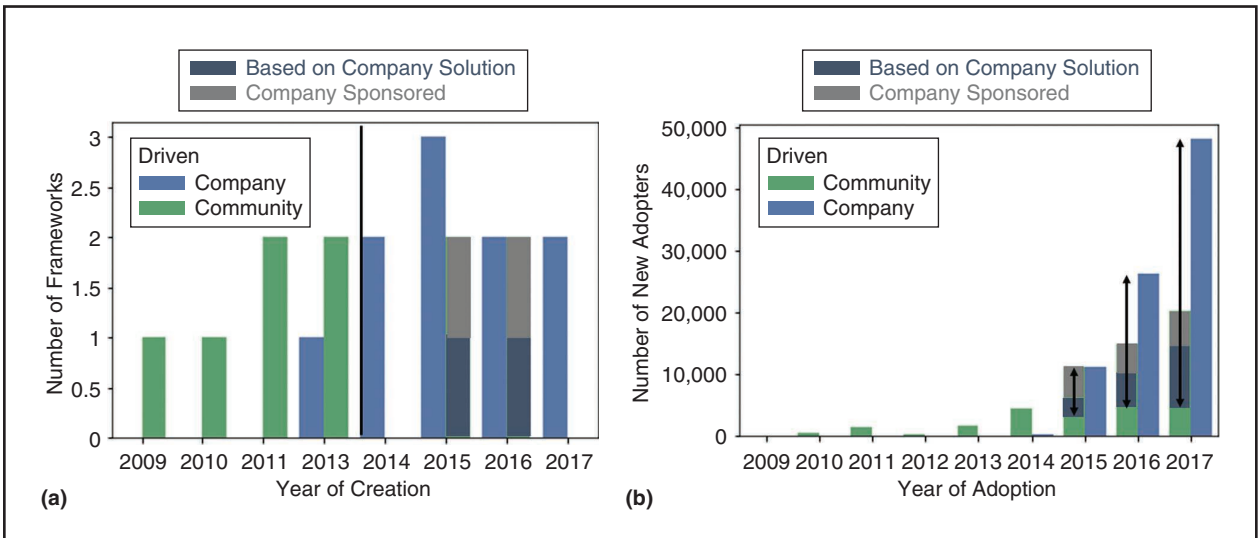


FIGURE 2. An analysis of community-driven ML frameworks based on company-driven solutions: (a) number of frameworks for each type of project per year of creation and (b) number of adopters for each type of project per year of adoption.

even more the discrepancy between the number of adopters of purely community-driven frameworks and frameworks with companies involved. In other words, in the last six years, companies have been dominating the development of ML technologies.

We also observed that company-driven ML frameworks are often deployed and optimized for commercial cloud offerings, which raises the

risks that these open source frameworks serve as baits to trap ML users and community developers into commercial platforms. It is extremely important to ensure that open source ML technologies remain accessible from diverse types of infrastructures, to prevent them from being biased toward the goals of specific organizations. Currently, the richest data sets reside within the servers of large

companies, hence more effort should be directed toward the development of open data sets, to empower a more diverse user base.

Skill specialization

Our analysis of the profile of contributors of the 20 top ML frameworks learned that, except for three frameworks, all ML development teams are hybrid. In company-driven ML

projects, professional researchers and engineers often contribute equally. The contribution of academic researchers is often limited to model design and training and they rarely contribute to the development or production of code. In contrast, in community-driven ML projects, the picture is significantly different, with professional and academic researchers writing almost the totality of the code.

Taking a closer look at the profile of professional researchers from either company-driven or community-driven frameworks, we observed that they typically hold a Ph.D. and have experience working in either an R&D lab or on an innovation product team, which signals that ML development is currently mostly driven by a few highly skilled professionals. Such a high entry bar is likely to have an adverse effect on innovation; it is important to develop tools and policies that enable the adoption of ML by a wider and more diverse user base, allowing them to express their creativity and achieve their use cases.

The two diversity challenges that we studied in open source ML frameworks, i.e., the dominance of companies and reliance on highly specialized personnel, are not necessarily new,^{3,7,11,12} yet the scale at which these ML software frameworks and projects are developing, as well as the (often safety-critical) domains in which they are applied, are cause for concern. Most intuitively, these challenges could lead to a shortage of qualified personnel, on top of the current shortage of open source contributors.¹³

At a higher level, the need to add specialized roles like industry researchers increases the need for

communication, while at the same time the dominance of companies makes the development process less transparent. A clear example of the latter is the “tensorflow-gardener”

ML frameworks backed by different companies, each with their own terminology and agenda. While the first issue could be addressed, at least to some extent, by including

For technologies that increasingly are controlling our lives, transparency of the development process is essential.

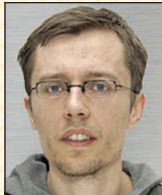
bot that recommits TensorFlow changes made internally by Google employees on the public GitHub repository, typically to hide employee names or to merge multiple commits at once. For technologies that increasingly are controlling our lives, transparency of the development process is essential. Finally, going back to Linus’ law,⁶ an overly biased, less-diverse contributor community also risks correlation with lower code quality, or at least higher costs to achieve a certain code quality.

So, what can be done about this? First of all, for the company dominance challenge, it is important to stress that we are not finger-pointing at the companies, since they have pushed many of the recent breakthroughs in ML/AI and also enabled democratization of those breakthroughs by open sourcing advanced frameworks like TensorFlow or PyTorch. In fact, they too are fighting for the (relatively) scarce AI talent,⁷ and would welcome any chance at attracting both competent researchers and engineers. Instead, the core issues seem to be 1) lack of AI training of potential contributors and 2) the proliferation of

AI courses in undergraduate curricula and organizing workshops and seminars on advanced AI technologies (which most universities and AI research centers have been doing the past couple of years), the second issue would require a standardization effort¹⁴ led by a neutral consortium. (Our study also considered a second measure of adoption, the 4,099 projects that used the ML frameworks. We refer to our paper for those results.¹⁰)

Both of these issues require long-term thinking. In the meantime, the company and/or high-tech bias in open source ML projects might be controlled by embracing more open development processes (e.g., MLOps)¹⁵ that stress communication between all stakeholders, including data scientists and industry researchers. Openness will likely spark more collaboration between different AI stakeholders and increase trust in AI technologies.

We believe that improving diversity of AI open source software projects, whether in terms of gender, race, geography, equity, or expertise, is essential to improve not only the quality⁶ of these projects, but also



BRAM ADAMS is an associate professor at Polytechnique Montréal, where he heads the Lab on Maintenance, Construction, and Intelligence of Software. Adams received a Ph.D. in computer science engineering from Ghent University, Belgium, in 2008. Contact him at bram.adams@polymtl.ca.



FOUTSE KHOMH is an associate professor at Polytechnique Montréal, where he leads the Software Analytics and Technology Lab. Khomh received a Ph.D. in computer science from the University of Montreal, Quebec, in 2010. Contact him at foutse.khomh@polymtl.ca.

the cohesion of their contributor communities¹⁶ and (ultimately) their sustainability. 🌱

References

1. S. M. West, M. Whittaker, and K. Crawford, "Discriminating systems: Gender, race and power in AI," AI Now Institute, Apr. 2019. Accessed on: Feb. 2020. [Online]. Available: <https://ainowinstitute.org/discriminating-systems.html>
2. T. Simonite, "AI is the future—But where are the women?" *Wired*, Aug. 17, 2018. [Online]. Available: <https://www.wired.com/story/artificial-intelligence-researchers-gender-imbalance/>
3. C. M. Melendez, "In AI, diversity is a business imperative," *Forbes*, Nov. 14, 2019. [Online]. Available: <https://www.forbes.com/sites/forbestechcouncil/2019/11/14/in-ai-diversity-is-a-business-imperative/#51c5aac8103f>
4. A. L. Hoffmann, "Data violence and how bad engineering choices can damage society," *Medium*, Apr. 30, 2018. [Online]. Available: <https://medium.com/s/story/data-violence-and-how-bad-engineering-choices-can-damage-society-39e44150e1d4>
5. IBM Research, "AI and bias." Accessed on: Feb. 2020. [Online]. Available: <https://www.research.ibm.com/5-in-5/ai-and-bias/>
6. E. S. Raymond and B. Young, *The Cathedral and the Bazaar: Musings on Linux and Open Source by an Accidental Revolutionary*, Sebastopol, California: O'Reilly Media, 2001.
7. J.-F. Gagné, G. Kiser, and Y. Mantha, "Global AI talent report 2019," Element AI. Accessed on: Feb. 2020. [Online]. Available: <https://jfgagne.ai/talent-2019>
8. Y. Shoham, R. Perrault, E. Brynjolfsson, J. Clark, J. Manyika, J. C. Niebles et al., "The AI index 2018 annual report," AI Index Steering Committee, Human-Centered AI Initiative, Stanford Univ., Stanford, CA, Dec. 2018. [Online]. Available: <http://cdn.aiindex.org/2018/AI%20Index%202018%20Annual%20Report.pdf>
9. K. Stathoulopoulos and J. Mateos-Garcia, "Gender diversity in AI research," Nesta, London, July 17, 2019. [Online]. Available: <https://www.nesta.org.uk/report/gender-diversity-ai>
10. H. B. Braiek, F. Khomh and B. Adams, "The open-closed principle of modern machine learning frameworks," in *Proc. 15th Int. Conf. on Mining Software Repositories (MSR)*, Gothenburg, Sweden, 2018, pp. 353–363.
11. E. Horvitz, "AI, people, and society," *Science*, vol. 357, no. 6346, pp. 7, July 7, 2017. doi: 10.1126/science.aao2466.
12. K. Crawford, "Artificial intelligence's white guy problem," *NY Times*, June 6, 2016. [Online]. Available: <https://www.nytimes.com/2016/06/26/opinion/sunday/artificial-intelligences-white-guy-problem.html>
13. The Linux Foundation, "The 2018 open source jobs report," 2018. [Online]. Available: <https://www.linuxfoundation.org/publications/2018/06/open-source-jobs-report-2018/>
14. C. E. Perez, "Why deep learning needs standards for industrialization," *Medium*, Feb. 9, 2019. [Online]. Available: <https://medium.com/intuitionmachine/challenges-for-ai-standardization-eab1de4fab0b>
15. N. Talagala, "Why MLOps (and not just ML) is your business' new competitive frontier," *Airtrends*, Jan. 30, 2018. [Online]. Available: <https://www.airtrends.com/machine-learning/ml-ops-not-just-ml-business-new-competitive-frontier>
16. G. Catolino, F. Palomba, D. A. Tamburri, A. Serebrenik, and F. Ferrucci, "Gender diversity and women in software teams: How do they affect community smells?" in *Proc. 41st Int. Conf. Software Engineering: Software Engineering in Society (ICSE-SEIS)*, May 2019, pp. 11–20.